

Question	Answer	Marks
7(a)	(a particle that) cannot be divided/subdivided (into smaller particles)	B1
7(b)(i)	$(p \times e) / (14u) = 4.1 \times 10^7$	C1
	$p = (4.1 \times 10^7 \times 14 \times 1.66 \times 10^{-27}) / (1.60 \times 10^{-19})$	C1
	$p = 6$ (answer should be an integer)	A1
7(b)(ii)	${}^{14}_6\text{X} \rightarrow {}^{14}_7\text{Z}$	B1
	${}^0_{-1}\text{e}^{(-)} \text{ or } {}^0_{-1}\beta^{(-)}$	B1
	${}^0_0\bar{\nu}_{(e)}$	B1
7(c)	(the nuclei are undergoing) β^+ decay	B1
	A correct explanation in terms of charge and a correct explanation in terms of energy Explanations in terms of charge: • (particles / decay) positively charged so cannot be β^- • (particles / decay) positively charged so could be / is β^+ • β^- (particles / decay) are negatively charged • β^+ (particles / decay) are positively charged Explanations in terms of energy: • range of energies so not α (particles / decay) • range of energies so is β (particles / decay) • β (particles / decay) have a range of energies • α (particles / decay) have discrete energies / not range of energies	B1